Group: Rohan Sampy, Maya Clapperton, Abrar Faruque, Jihyun Bang, Brenda Soljic

Molecular Pathogenesis Assignment

Please pick an inherited disease from the assigned list and answer the following questions.

Disease Name: Alagille Syndrome

Disease Phenotype:
Complicated presentation with multiple organ involvement. Pathology in the liver is most common (cholestasis), pulmonary valve stenosis, renal tubule defects, extra-cardiac vascular defects, bone deformities, characteristic facies.[1]

Inheritance Pattern most involved:
Autosomal dominant (with variable penetrance and other factors that are not well understood that affect expressivity).[1-4]

Gene Name and Acronym of gene most involved in the disorder:
JAG1, Notch2.[1-4] We will focus on JAG1 here because it is more commonly mutated, is associated with higher morbidity, and is the most well-understood gene involved in pathogenicity.

Type of mutation/s found in the gene (e.g. missense, nonsense etc):
Missense (more common for NOTCH2), nonsense (more common for JAG1), full deletion, Pathogenic SNPs, frameshift.[2]

What is the pathogenesis based on genetic information? (i.e. LOF, GOF etc)?
Loss of function[1,2]

What is the evidence to support the pathogenesis?
Mouse-model work has shown that these genes are involved, and knockout studies in a developing mouse embryo show that LOF during development leads to Alagille-like syndrome.[1] Functional studies have shown that mutated JAG1 stays intracellular and does not reach the cell membrane where it can bind to NOTCH2.[1]

Take one mutation and describe what is happening on the molecular level as to how the mutation is affecting the RNA or protein. If no one is known, what is a hypothesis for the pathogenic mechanism?
The c.2066del results in a deletion of cytosine at position 2066 in exon 13, creating a frameshift mutation and a premature stop codon. This results in a truncated JAG1 transcript and non-functional protein. Some EGF domains that bind to NOTCH2 are lost, and binding to NOTCH2/signaling does not occur (UCSC Genome Browser).

If known, describe how mutations in this gene are altering cellular function to cause the specific phenotype of the disorder. If not known, what is the hypothesis as to how the gene is altering cell function in causing the phenotype?
In Alagille syndrome, pathogenic variants in JAG1 (the Notch ligand) or NOTCH2 (the Notch2 receptor) are thought to cause disease primarily by reducing Notch signaling, since Jagged1–Notch2 binding normally triggers Notch2 cleavage and nuclear translocation of its intracellular domain to regulate transcriptional programs during development.[2]

Most JAG1 variants are loss-of-function and act via haploinsufficiency, meaning insufficient ligand dosage leads to inadequate signaling across key cell–cell interfaces.[4] Mechanistically, animal work supports a developmental biliary explanation: Jag1 expression in the portal vein mesenchyme is required for ductal plate remodeling into tubular intrahepatic bile ducts, so reduced signaling plausibly yields bile duct paucity and downstream cholestasis, with secondary consequences like impaired bile/lipid handling (e.g., hypercholesterolemia and xanthomas).[1]

For NOTCH2-associated disease, ALGS is still framed as a Notch signaling disorder, but the variant-level functional consequences are less well understood, making the exact cellular mechanism harder to pin down beyond a likely Notch signaling deficiency.[4]

Finally, the variable expressivity (even among relatives with the same pathogenic variant) is the reason the field leans on a modifier-gene/second-hit hypothesis, where other genetic factors outside JAG1/NOTCH2 tune residual pathway activity and thereby shift organ involvement and severity.[1-4]

Please list your references here. Though you can refer to websites and review papers, they should always refer to primary references which you should also use here.

1. Kohut, T. J., Gilbert, M. A. & Loomes, K. M. Alagille Syndrome: A Focused Review on Clinical Features, Genetics, and Treatment. *Semin Liver Dis* 41, 525–537 (2021).
2. Li, L. *et al.* Alagille syndrome is caused by mutations in human Jagged1, which encodes a ligand for Notch1. *Nat Genet* **16**, 243–251 (1997).
3. Ranchin, B., Meaux, M.-N., Freppel, M., Ruiz, M. & De Mul, A. Kidney and vascular involvement in Alagille syndrome. *Pediatr Nephrol* **40**, 891–899 (2025).
4. Vandriel, S. M. *et al.* Phenotypic Divergence of *JAG1* - and *NOTCH2* -Associated Alagille Syndrome & Disease-Specific *NOTCH2* Variant Classification Guidelines. *Liver International* **45**, e70251 (2025).